FLEET & SHORE PERSONNEL TRAINING

Engine Resource Management Training

Why engine room incidents trace back to team coordination as often as technical failure — and the ERM skills that keep the engine room safe and ready.

STCW Table A-III/1

IMO ERM Model Course 2.07

Company SMS

COURSE MAP

What We Will Cover

01 **Understanding ERM**
Human and team factors behind engine room incidents

02 **Engine Room Team Roles**
Who does what in the engine room hierarchy

03 **Communication & Watch Handover**
Clear exchanges and a disciplined handover every time

04 **Standing Orders & Watchkeeping Discipline**
Turning the Chief's orders into daily practice

05 **Managing Abnormal Situations & Alarms**
Staying ahead of the alarm, not behind it

06 **Fatigue & Human Factors**
Managing human limits in a demanding environment

07 **Learning From Failure**
Well-documented ERM failures and their lessons

SECTION 1

Understanding ERM

01

Why engine room incidents so often trace back to team coordination and communication, not a one-off technical fault.

WHY ERM EXISTS

Human Error in the Engine Room

Investigations into engine room casualties point to the same team failures seen on the bridge.

- Marine casualty investigations regularly find that engine room incidents involve missed communication, unclear roles or poor monitoring as often as a genuine technical fault.
- Engine Resource Management is the set of skills that make full use of all available resources — people, procedures, alarms and equipment — to run the engine room safely.
- Like BRM, ERM was adapted from Crew Resource Management principles developed in aviation, applied to the particular pressures of the machinery space.
- A technically excellent engineer can still contribute to an unsafe engine room if information is not shared, verified and acted on as a team.
- STCW Table A-III/1 and A-III/2 require ERM as a mandatory competence for engineer officers, assessed alongside technical watchkeeping skills.

Not Just the Chief

ERM distributes responsibility for safety across the whole engine room team — every watchkeeper and rating has a duty to notice, question and speak up.

CORE SKILLS

The Four Pillars of Engine Resource Management

The same four pillars used on the bridge apply directly to the machinery space.

Situational Awareness

Keeping an accurate, current picture of every running system, its parameters and its normal behaviour.

Communication

Clear, closed-loop exchange of information between every member of the engine room team.

Workload Management

Distributing tasks and monitoring duties so no one watchkeeper is overloaded during an evolution.

Leadership & Assertiveness

Directing the team clearly while remaining genuinely open to being challenged by any member.

WHY IT BECAME MANDATORY

How ERM Entered Standard Practice

ERM followed a similar path to BRM, driven by real casualties rather than theory.

1980s-90s

A run of serious engine room casualties, on otherwise technically sound ships, drove the industry to formalise ERM training

STCW

ERM competence is now examined and assessed alongside watchkeeping and technical engineering skills

SMS

Most company Safety Management Systems now build ERM principles directly into engine room procedures and checklists

SECTION 2

02

Engine Room Team Roles

Who does what in a well-run engine room, and how manning should scale as the task gets riskier.

DEFINING ROLES

Who Does What in the Engine Room

Clear roles, agreed before a critical evolution begins, are one of the simplest ERM tools available.

- The Chief Engineer holds overall responsibility for the safe and efficient operation of all machinery, and for the competence and organisation of the engine department.
- The Engineer Officer of the Watch (EOOW) is responsible for the safe operation and monitoring of machinery during the watch, following the Chief's standing orders.
- Watchkeeping ratings support the EOOW with rounds, monitoring and immediate hands-on response, and must know when and how to raise a concern.
- During manoeuvring, bunkering, or maintenance on running equipment, additional personnel are added and roles are explicitly briefed beforehand.
- Every engine room role should be assigned and understood before a critical evolution begins, not assumed or left to whoever happens to be present.

Manning Scales With Risk

Engine room manning should increase for manoeuvring stand-by, bunkering and any work on running machinery — not stay fixed regardless of task complexity.

ROLE CLARITY

Assigning and Respecting Roles

Ambiguity about who is monitoring what is one of the fastest ways an engine room's safety margin quietly disappears.

Do

- Brief who is monitoring which system, who is on rounds and who is on standby before a critical evolution
- Scale up manning proactively for manoeuvring, bunkering or high-risk maintenance
- Confirm handover of watch responsibility verbally and explicitly, never by assumption
- Keep the EOOW's monitoring role free of unrelated administrative tasks during critical periods

Don't

- Don't let one watchkeeper monitor multiple critical systems alone during a high-risk evolution
- Don't assume a rating understands their task without a stated brief
- Don't let rank alone decide who checks whom — every team member must be able to raise a concern
- Don't leave the engine control room unattended during manoeuvring stand-by or cargo operations

MANNING LEVELS

Matching Team Size to Task

Engine room manning is a deliberate decision, reviewed as the situation changes, not a fixed roster.

Situation	Minimum Team	Why
Sea passage, steady state, good conditions	EOOW + rating on call	Standard watchkeeping provides an adequate safety margin
Manoeuvring stand-by or restricted waters	Chief Engineer + EOOW + additional engineer + rating	Full team needed for rapid response to bridge orders
Bunkering or transfer operations	Duty engineer + dedicated watchman at each connection	Continuous monitoring required to prevent overflow or spill

SECTION 3

03

Communication & Watch Handover

Clear exchanges and a disciplined handover, every watch, without exception.

WATCH HANDOVER

Why Handover Discipline Matters

Many engine room incidents that unfold hours into a watch can be traced back to a rushed handover at its start.

- A rushed or incomplete handover is one of the most common starting points for an engine room incident that unfolds hours into the next watch.
- The outgoing EOOW must brief the incoming officer on running machinery status, any equipment out of service, ongoing work and any abnormal readings.
- A structured handover checklist — standing orders, bunker/tank levels, maintenance in progress, alarms silenced or overridden — should be worked through, not recited from memory.
- The incoming EOOW should personally verify key parameters and system status before accepting the watch, not rely solely on the outgoing officer's word.
- No watch should be handed over during an unresolved abnormal situation or alarm condition without both officers explicitly agreeing on the plan going forward.

Never a Silent Handover

A handover completed in silence, with no questions asked, is a warning sign in itself — a good handover generates a short conversation, not just a signature.

COMMUNICATION TOOLS

Formal Channels the Engine Room Relies On

Structured exchanges catch what casual conversation reliably misses.

Watch Handover Checklist

A structured, checklist-supported exchange of machinery status between outgoing and incoming watchkeepers.

Engine Order Telegraph / Bridge Link

The formal channel for manoeuvring orders between bridge and engine room, always acknowledged and read back.

Standing & Night Orders

The Chief Engineer's written instructions defining when the EOOW must call the Chief, kept current and briefed to every engineer.

Permit-to-Work Briefing

A structured briefing before any maintenance on running or isolated equipment, confirming isolation, PPE and emergency arrangements.

COMMUNICATION DISCIPLINE

Closed-Loop Communication in the Engine Room

Noise, heat and distance make closed-loop discipline even more important than on the bridge.

Do

- Acknowledge and read back every order received from the bridge or the Chief Engineer
- Report abnormal readings immediately, even if you suspect they will resolve themselves
- Use standard terminology for equipment and systems so nothing is ambiguous
- Confirm out loud once an action has been completed, not just marked on a log

Don't

- Don't accept a mumbled or partial acknowledgement as confirmation of an order
- Don't let noise, PPE or distance excuse an unclear or unconfirmed instruction
- Don't wait for a supervisor to ask before reporting a developing problem
- Don't rely on a single log entry in place of a verbal handover conversation

SECTION 4

Standing Orders & Watchkeeping Discipline

04

Turning the Chief Engineer's written orders into consistent, daily watchkeeping practice.

STANDING ORDERS

Turning Orders Into Daily Practice

Standing orders only work if they are read, understood and genuinely followed on every watch.

- Standing orders set out the Chief Engineer's fixed requirements for every watch — parameters to monitor, actions on specific alarms, and when the Chief must be called.
- Night orders add instructions specific to the coming hours: planned maintenance, expected manoeuvring, or equipment recently returned to service.
- Every engineer must read and sign standing and night orders at the start of each watch, and treat them as a requirement, not a formality.
- Watchkeeping discipline includes regular rounds of the machinery spaces, not only monitoring from the control room — abnormal sounds, smells and vibration are often caught on rounds first.
- Deviating from a standing order, even briefly, should be discussed with the Chief Engineer, not decided alone in the moment.

Read, Don't Skim

Standing and night orders are only as useful as the attention given to them — a signature without reading them defeats the entire purpose of the system.

DISCIPLINE

Watchkeeping Discipline in Practice

The habits that separate a genuinely well-run watch from one that only looks fine on the log.

Do

- Complete engine room rounds on schedule, even when everything appears normal on the panel
- Log parameters and observations promptly and accurately, not retrospectively from memory
- Call the Chief Engineer whenever standing orders require it, without hesitation
- Keep the engine control room manned and monitored throughout the watch

Don't

- Don't silence or acknowledge an alarm without identifying its actual cause
- Don't skip a scheduled round because the watch has been busy with other tasks
- Don't treat standing orders as guidance that can be quietly set aside under pressure
- Don't leave abnormal readings unlogged because they returned to normal on their own

DAILY DISCIPLINE

What Good Watchkeeping Looks Like

Three simple habits, done consistently, account for most of the difference between watches.

Rounds

Scheduled physical rounds of the machinery spaces, not only remote monitoring from the control room

Log

Accurate, contemporaneous logging of parameters, actions taken and any abnormal readings

Call

Calling the Chief Engineer promptly whenever standing orders or judgement say it is required

SECTION 5

Managing Abnormal Situations & Alarms

05

Staying ahead of a developing problem, not just responding to the alarm once it finally sounds.

Staying Ahead of the Alarm

By the time an alarm activates, the problem has usually been developing for some time.

- An alarm is the last stage of a developing problem, not the first — trending parameters correctly should give warning well before an alarm activates.
- On any alarm, the immediate priority is to identify the actual cause, not simply to silence the audible signal and move on.
- A single abnormal reading should prompt a wider check of related systems, since engine room systems are rarely independent of one another.
- Multiple simultaneous alarms can indicate a common root cause — a power failure or a shared cooling system fault — rather than several unrelated faults.
- Every abnormal situation, resolved or not, should be reported to the EOOW or Chief Engineer and logged, so the picture is never held by one person alone.

Don't Silence and Forget

Silencing an alarm stops the noise, not the problem — treat every alarm as requiring a specific, closed-out cause before it is considered resolved.

RESPONSE DISCIPLINE

Responding to a Developing Problem

A calm, structured response is what keeps a developing problem from becoming a serious incident.

Do

- Identify the actual cause of an alarm before silencing it and moving on
- Check related systems whenever one abnormal parameter is found
- Inform the EOOW or Chief Engineer of any abnormal situation, however minor it seems
- Follow the emergency procedure or checklist for the specific alarm, not memory alone

Don't

- Don't silence an alarm without understanding what caused it
- Don't assume an intermittent alarm will not recur once it has stopped
- Don't attempt an unfamiliar repair on running machinery without informing the watch team
- Don't delay calling for help because the situation might still resolve on its own

DIAGNOSIS

Reading Multiple Alarms Correctly

The pattern of alarms is often as informative as any single alarm on its own.

Pattern	Likely Meaning	First Action
Several unrelated alarms at once	Possible power supply or common sensor bus fault	Check main power and common systems before individual equipment
One system's parameters drifting steadily	Developing mechanical or process fault	Trend the reading, check related systems, inform the EOOW
Alarm clears itself after silencing	Underlying cause may still be present	Investigate and log the cause, do not assume it has resolved

SECTION 6

06

Fatigue & Human Factors

Managing the human limits — heat, noise, sleep debt — that shape performance in a demanding environment.

FATIGUE

How Fatigue Degrades Engine Room Performance

The engine room adds its own physical stressors on top of the fatigue every watchkeeper already carries.

- Heat, noise and vibration in the engine room add physical fatigue on top of the sleep debt common to shipboard rest patterns.
- Fatigue impairs the ability to notice a slowly developing abnormal trend well before it impairs obviously physical tasks.
- STCW rest-hour requirements set minimum rest periods, but paperwork compliance does not guarantee a watchkeeper is genuinely fit for duty.
- A fatigued engineer is more likely to silence an alarm without full investigation, skip a round, or miss a subtle change in a familiar sound or reading.
- Reporting fatigue honestly, and adjusting watch schedules or manning in response, is a Company SMS responsibility as much as an individual one.

Heat Adds Up

Engine room heat stress accelerates fatigue faster than equivalent hours on the bridge — build in relief and hydration breaks on long or hot watches.

What Affects Engine Room Safety Beyond Skill

Technical competence alone does not protect against these everyday pressures.

Noise & Communication

High ambient noise makes verbal communication harder and increases the risk of a missed or misheard instruction.

Heat Stress

Sustained heat exposure degrades concentration, reaction time and decision-making well before it feels dangerous.

Confined & Repetitive Tasks

Working in confined spaces on repetitive tasks can narrow attention and make it easy to miss a developing abnormal sign.

Workload Peaks

Manoeuvring, bunkering and maintenance can converge, overloading a small watch unless planned for in advance.

The Numbers Behind Rest Requirements

STCW rest hours are a legal floor, not a target — good practice builds in margin above the minimum.

10 hrs/24

Minimum STCW rest requirement in any 24-hour period, split into no more than two periods

77 hrs/7d

Minimum STCW rest requirement in any 7-day period — a floor, not a target to aim for

Heat

Scheduled relief and hydration breaks should increase automatically as engine room ambient temperature rises

SECTION 7

Learning From Failure

07

Well-documented ERM failures from real casualties, and the lessons every engine room team should carry forward.

Why These Cases Are Still Taught

The same pattern of human and organisational failure appears across very different types of ship.

- Engine room casualty investigations, like bridge ones, repeatedly find teamwork and communication failures behind incidents that were technically preventable.
- A well-maintained engine room can still suffer a serious incident if an alarm is silenced without investigation, or a handover leaves a hazard unmentioned.
- These cases are taught industry-wide specifically because the failure was human and organisational, not a one-off mechanical surprise.
- Studying a real casualty in a classroom is far cheaper than the ship, its crew or the Company learning the same lesson at sea.
- The purpose of reviewing these cases is not to assign blame to individuals, but to recognise the same warning signs before they repeat.

No Blame, Real Lessons

The Company's SMS reporting culture exists so that these lessons reach the whole fleet without punishing the people who report them.

CASE STUDIES

Well-Documented ERM Failures in Real Casualties

These widely studied cases are still taught across the industry because the failure was in teamwork, not seamanship.

Casualty	ERM Failure Identified	Lesson for the Engine Room Team
Container ship engine room fire, 2006	An alarm was silenced repeatedly without the underlying fault being investigated	Every alarm requires an identified, closed-out cause before it is considered resolved
Cruise ship blackout and grounding, 2006	A known fault was not clearly handed over between watches, and alarms were not acted on in time	Structured handovers and prompt escalation of developing electrical faults
Bulk carrier engine room flooding, 2015	A valve left open during maintenance was not communicated to the returning watch	Positive confirmation of system status before returning equipment to service

WHAT THE COMPANY EXPECTS

Turning Lessons Into Practice

ERM is not a certificate earned once; it is a daily discipline the whole fleet is expected to practise.

Brief

Every watch and every critical evolution begins with a clear team briefing of roles, plan and known risks

Challenge

Every team member is expected, and protected, to raise a concern about any abnormal reading or unclear instruction

Report

Near-misses and ERM concerns are reported through the SMS without blame, so lessons reach the whole fleet

COURSE COMPLETE

A well-drilled engine room team, working to clear standards and open communication, is what keeps machinery — and the people around it — safe.

Company DPA — 24/7 emergency line

Fleet Technical Department — mng@gloryship.com.tr

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